

RESOURCE-EFFICIENT PRIVACY-FIRST WEARABLE FALL DETECTION

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MOTIVATION

The development of wearable fall detection systems is critical for elderly safety, but traditional architectures rely on transmitting raw, highly sensitive biometric data to centralized servers.

- **The Privacy Vulnerability:** Centralizing data exposes intimate patient health patterns and daily habits to significant security risks and potential misuse.
- **The Hardware Bottleneck:** Securing this data with Homomorphic Encryption and Differential Privacy typically requires massive computational overhead, which easily overwhelms the battery and memory limits of standard edge wearables.
- **Our Secure Edge Solution:** We engineered a privacy-first Federated Learning framework. By keeping all raw data strictly on-device and encrypting model updates, we guarantee patient privacy without sacrificing the clinical accuracy required to detect life-threatening falls.

EXPERIMENT DESIGN

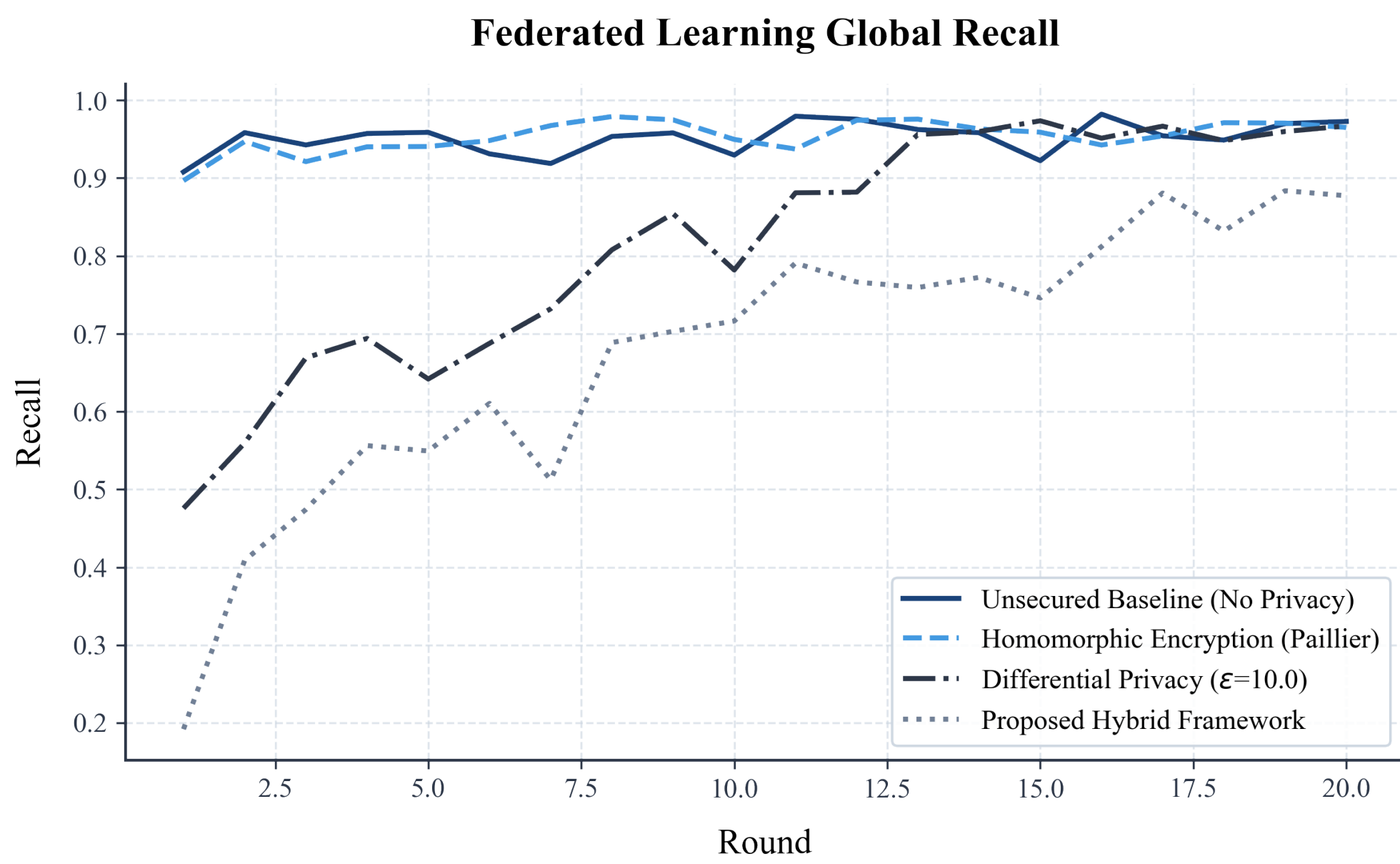
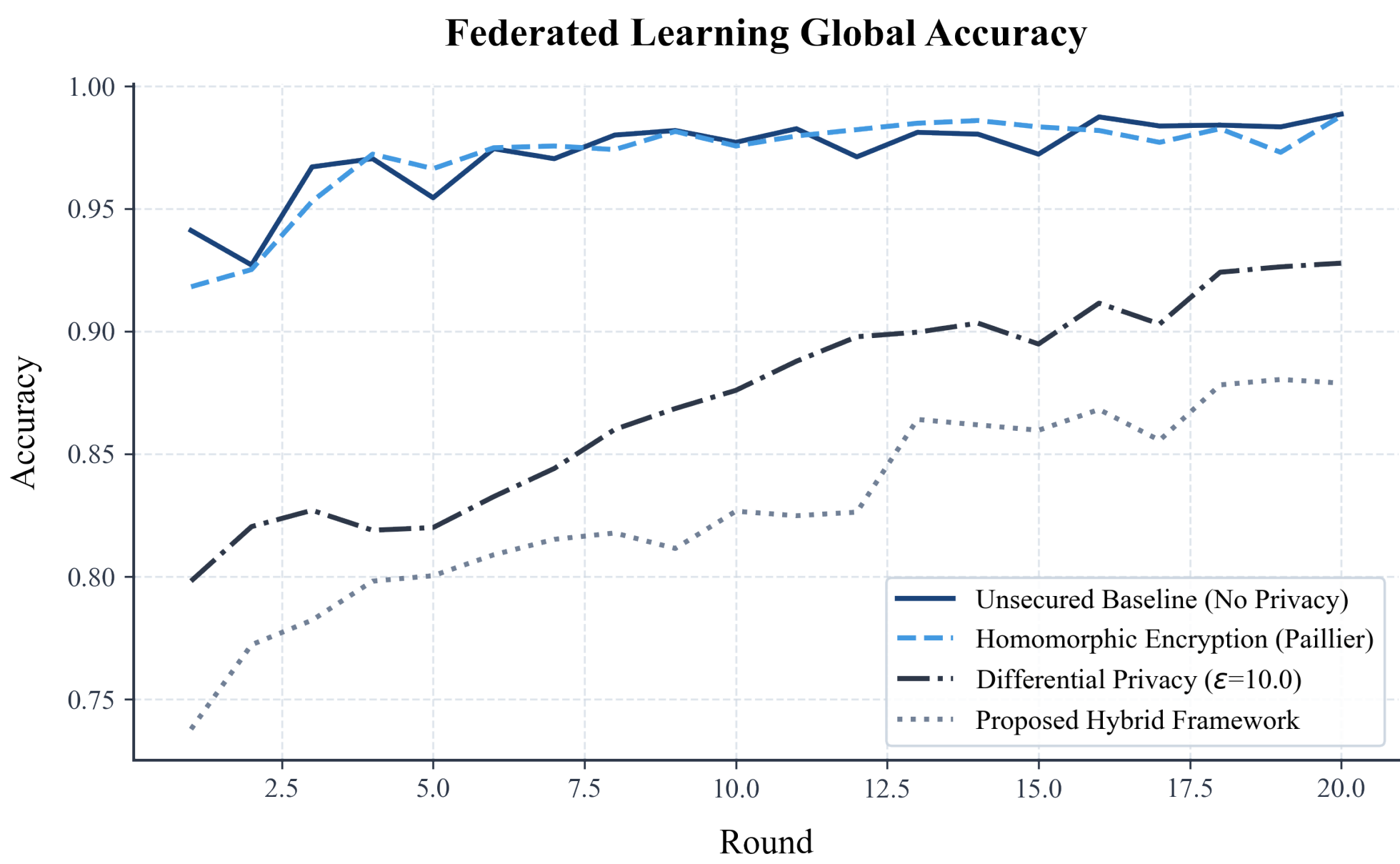
To balance the heavy computational tax of Homomorphic Encryption with the clinical necessity of high accuracy, we rigorously optimized the entire machine learning pipeline for low-power edge devices:

- **Data Pipeline:** Utilizing the **SisFall** dataset, accelerometer readings were downsampled to **50 Hz** and segmented into **2.56-second windows** to aggressively reduce the memory footprint.
- **Ultra-Lightweight Model:** A custom 1D-CNN was engineered with exactly **562 parameters**. This micro-architecture drastically minimizes the encrypted payload size.
- **Privacy Measures:** **Differential Privacy** ($\epsilon = 10.0$) was applied to prevent the model from memorizing individual subjects, alongside **Paillier Homomorphic Encryption** to securely aggregate all transmitted updates.

For complete implementation details, refer to the project repository (see QR code).

FEDERATED MODEL CONVERGENCE & PRIVACY TRADE-OFFS

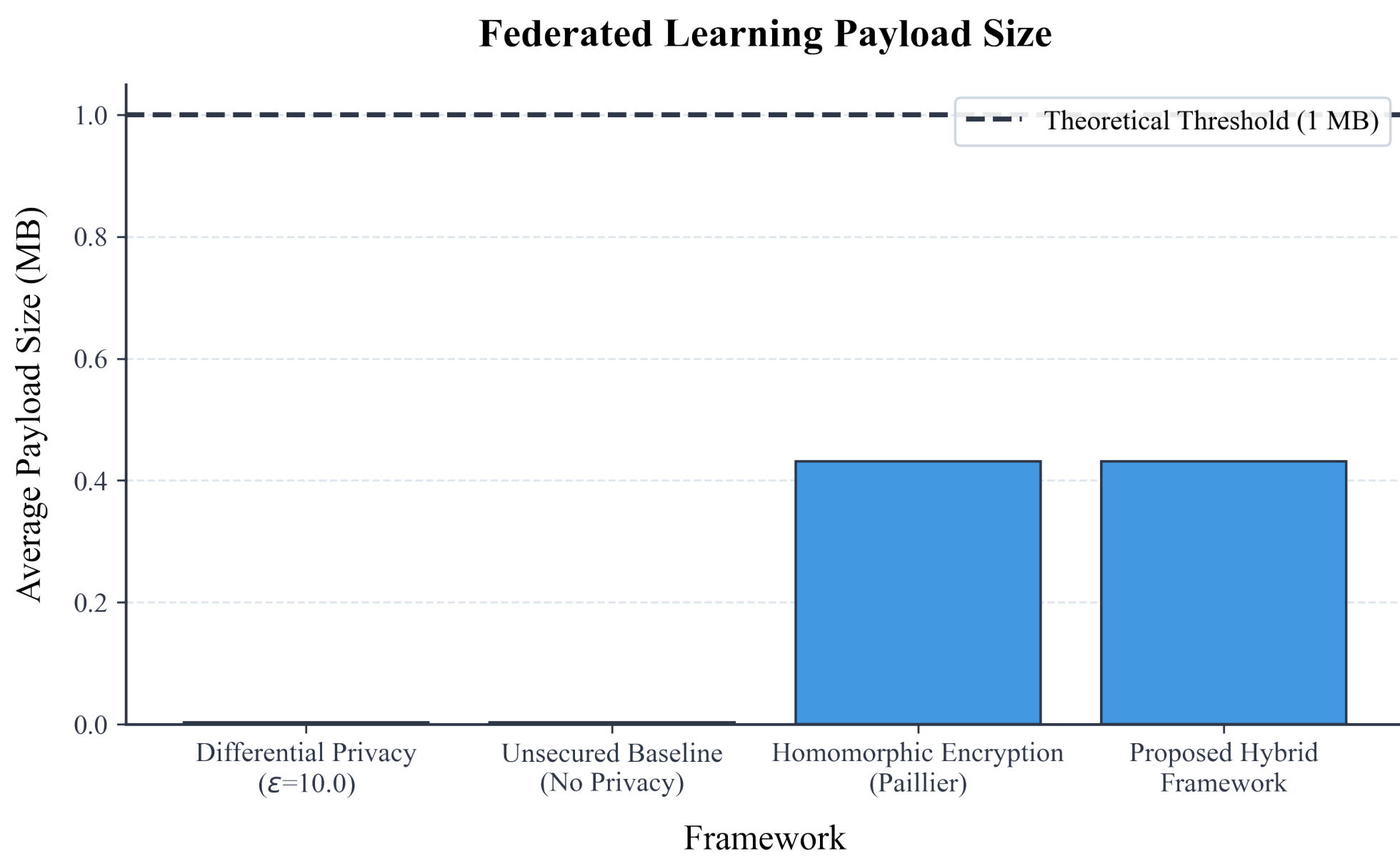
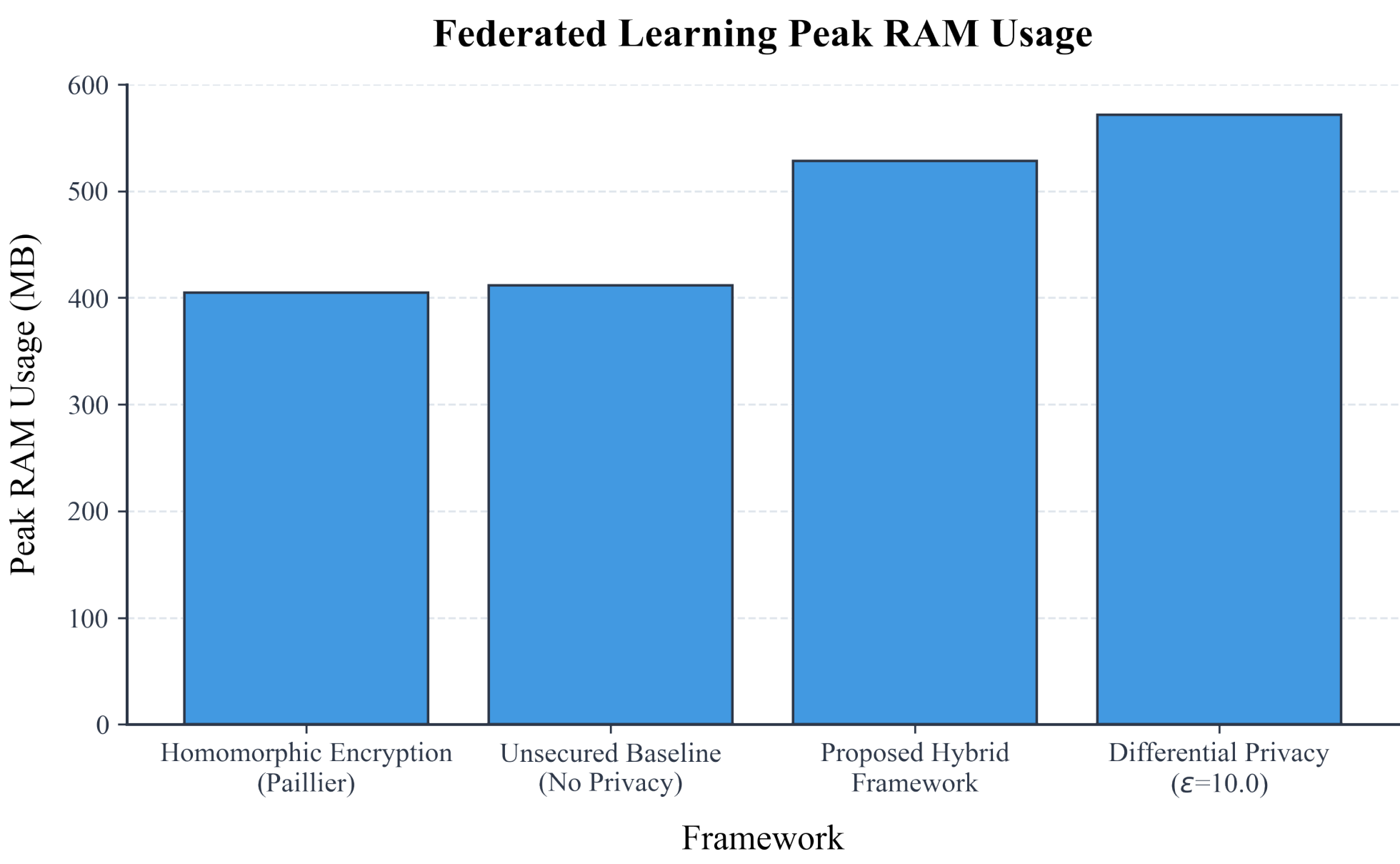
Global accuracy and recall were measured over 20 communication rounds to evaluate model convergence under privacy constraints.



As theoretically expected, the hybrid framework exhibits a modest drop in performance compared to the other frameworks due to the compounded complexity of privacy noise and encryption limits. However, converging to nearly **90% accuracy and recall** under industry-standard security proves the framework successfully preserves the essential clinical utility required for reliable fall detection.

COMPUTATIONAL COST & EDGE DEVICE VIABILITY

The computational footprint was assessed during simulated training to ensure the framework's feasibility for low-power wearables.



Notably, integrating heavy differential privacy and homomorphic encryption introduces minimal overhead; the hybrid framework's peak RAM usage remains strictly comparable to the unsecured baseline. Furthermore, the ultra-lightweight 1D-CNN restricts the **encrypted payload to just 0.4 MB**, staying comfortably below the 1.0 MB threshold for efficient Bluetooth (BLE) transmission.

HYBRID PRIVACY-PRESERVING FEDERATED FRAMEWORK (HPPFL)

HYBRID PRIVACY-PRESERVING FEDERATED LEARNING (HPPFL) FRAMEWORK

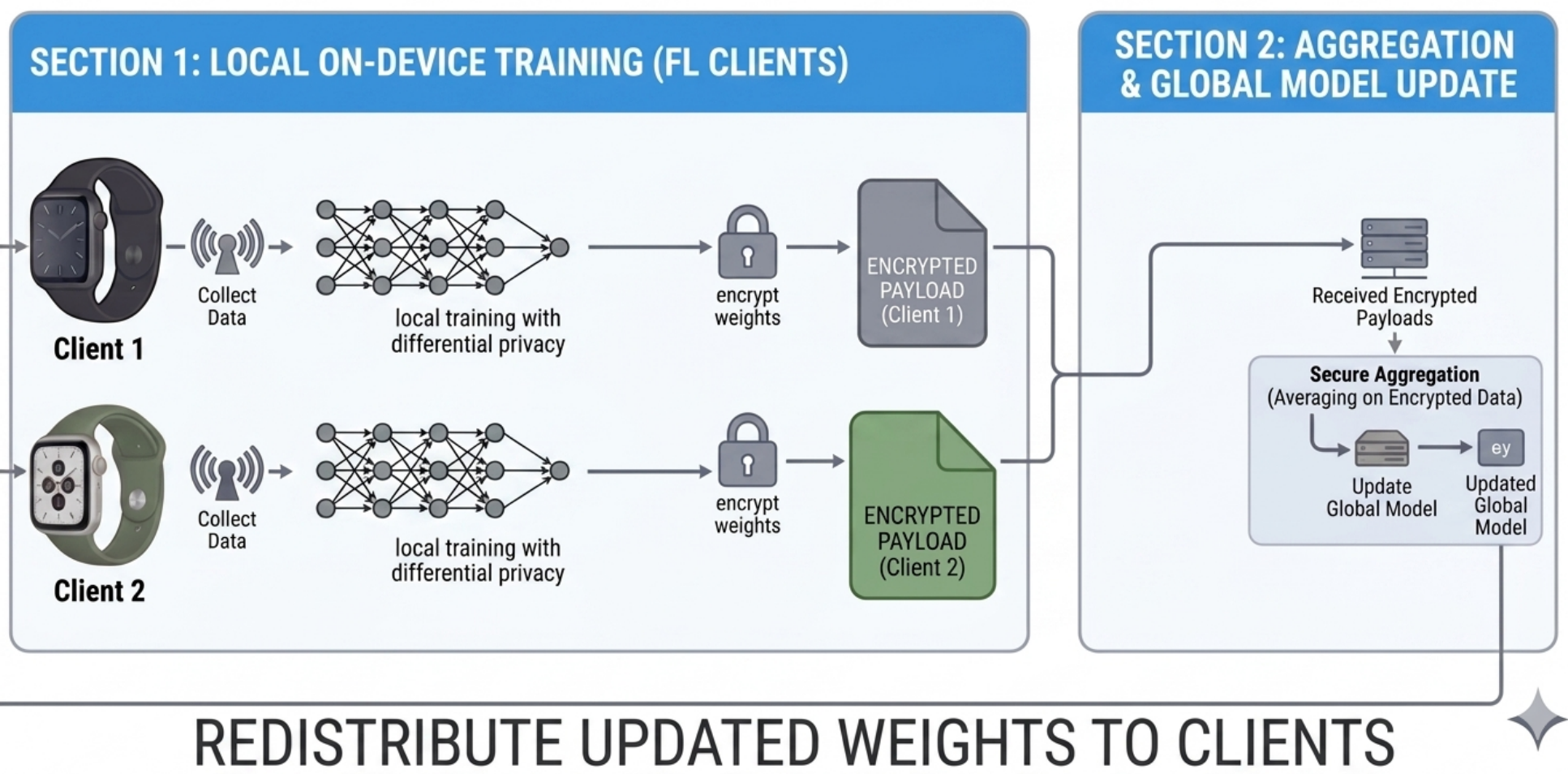


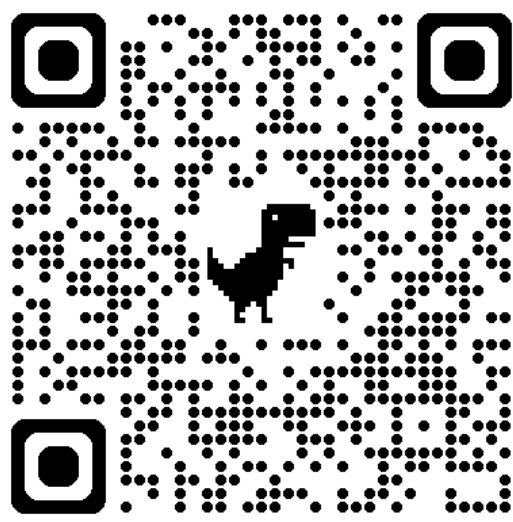
Image Created with Generative AI

The **HPPFL framework** is built on three core pillars:

- **Local Privacy:** Raw data never leaves the device, securing user information at the source.
- **Efficient Security:** Lightweight models train with **differential privacy** to obfuscate patterns before transmission.
- **Zero-Trust Aggregation:** The server updates global weights strictly using **homomorphically encrypted** payloads.

PROJECT REPOSITORY

Scan to explore the full HPPFL codebase, including the 1D-CNN architecture and federated learning simulation scripts on GitHub.



FUTURE CONSIDERATIONS

Alternative Cryptosystems

Investigate advanced or hybrid encryption schemes, such as PASTA, to further reduce communication overhead and computational latency on edge devices.

Broader Applications

Evaluate the framework's adaptability to other sensitive, time-series edge computing problems, such as continuous cardiac or seizure monitoring.

Model Interpretability

Integrate SHAP (SHapley Additive exPlanations) to enhance clinical trust by making the ultra-lightweight 1D-CNN's predictions fully interpretable.